

Round 3 Analytical Report

[Topic: Public Reaction to a Major Cyber Security Incident]

by Team Tekton – Yukta H, Chaaru Shree S

1. Introduction

The rapid spread of information through social media and online news platforms has changed how public reactions to major cybersecurity incidents develop and evolve. When a cybersecurity incident affects a large organization, online discussions can reflect changes in public sentiment, engagement, operational concerns, financial implications, and emerging issues over time.

This report analyzes the public reaction to the Boston Scientific cybersecurity incident using a self-collected dataset of posts and news items gathered during the Round 3 analysis period. Data from X (Twitter) and Google News were combined and analyzed using the NLP models developed during Round 2. The analysis examines sentiment, user engagement, discussion topics, key entities, temporal patterns, and changes in the nature of the discussion across different stages of the incident.

The objective is to move beyond a static classification of individual posts and identify how the online conversation changed over time, including sentiment shifts, engagement spikes, emerging technical or service-related concerns, and increasing attention to financial impact. The findings provide a structured view of the evolving public reaction to the cybersecurity incident based on the collected data and model predictions.

2. Background of the cybersecurity incident considered

2.1 Incident overview

The incident considered in this study is a cybersecurity incident involving Boston Scientific, a medical technology company.

The incident affected the company's IT systems and disrupted some of its business operations.

2.2 Operational impact

The collected data includes reports of disruptions to:

- business operations
- order processing
- manufacturing
- shipment of medical devices

The dataset also contains discussions about medical devices and services, including pacemakers, transmitters and remote monitoring.

2.3 Financial impact

Later discussions in the collected data focused more on the financial effects of the incident. These included:

- sales
- financial guidance
- company forecasts
- stock performance

2.4 Relevance to this study

This incident was selected because it provides a clear event for studying how online public discussion changes over time.

The analysis therefore examines changes in sentiment, engagement, topics and key entities across the collected posts and news items.

3. Objectives

The objectives of this analysis are:

- To collect and prepare relevant data from X (Twitter) and Google News related to the Boston Scientific cybersecurity incident.
- To clean, combine and validate the collected data while maintaining the required dataset structure.
- To apply the sentiment classification model developed in Round 2 to the collected data and examine sentiment patterns over time.
- To apply the Round 2 topic classification model to identify the predicted discussion categories in the collected data.
- To analyse activity and engagement patterns in the X posts, including likes, comments and shares, and identify major engagement spikes.
- To identify the main topics, themes and entities appearing in the collected posts and news items.
- To compare discussion patterns across important dates and identify changes in sentiment, activity and thematic focus.
- To examine the major changes observed in the conversation and relate them to the events or updates represented in the collected data.
- To present the findings through time-based analysis and visualisations, while clearly distinguishing model predictions and observed patterns from causal claims.

4. Data collection method

4.1 X (Twitter) data collection

X (Twitter) data was collected using XPorter, a Chrome extension that exports posts from X into structured files such as CSV. The data was collected from relevant X search results for the Boston Scientific cybersecurity incident during the selected analysis period.

The exported X data contained fields such as post ID, post text, timestamp, likes, replies and reposts. These fields were later mapped to the required Round 3 dataset structure.

XPorter repository: <https://github.com/Lemelson/xporter-extension>

4.2 Google News data collection

News data was collected programmatically using the Google News RSS feed for the search query: "Boston Scientific" cyberattack

A Python script, news_collection.py, was used to retrieve and process the RSS feed.

The main Python libraries used were:

- requests — used to send the request to the Google News RSS feed.
- feedparser — used to parse the RSS feed and extract the available article information.
- pandas — used to organise the extracted records into a structured dataset.

The collected news records included the article title/text and publication timestamp. Engagement fields such as likes, comments and shares were not available from the Google News RSS feed and were therefore kept as missing values rather than being replaced with zero.

4.3 Data preparation and combination

The X and Google News datasets were processed using the Python script `prepare_round3_data.py`.

The script:

- loaded the X and news datasets,
- standardised their column names,
- mapped the available fields to the required Round 3 schema,
- combined the two sources,
- converted and standardised timestamps,
- sorted the records chronologically,
- checked for duplicate text IDs, and
- saved the combined dataset for further analysis.

The final dataset used for the analysis contains the following nine fields: `text_id`, `source`, `post_text`, `timestamp`, `likes`, `comments`, `shares`, `sentiment_label`, `topic_category`

The sentiment and topic fields were subsequently generated using the NLP models developed in Round 2.

5. Time window

5.1 Real-world incident time window

The cybersecurity incident involving Boston Scientific was identified on August 25, 2026. The incident affected the company's information technology systems and caused disruption to business operations, including manufacturing, order processing and shipping.

The incident and its operational impact continued into the following weeks, with updates on service restoration and operational recovery appearing during the collected period. The company later reported that manufacturing, order fulfillment and shipping operations had been restored.

Therefore, the real-world incident timeline considered in this study covers the period from the initial identification of the incident on August 25, 2026, through the subsequent recovery and operational updates.

5.2 Data collection time window

The final dataset covers the period from August 26, 2026 to September 20, 2026.

This window was selected to capture the online discussion surrounding the incident, including the initial reaction, operational disruption, subsequent updates and later discussion about the impact of the incident.

The final dataset contains 190 records from two sources:

- 97 news records
- 93 X (Twitter) records

The daily record distribution is shown below.

Date	News	Twitter	Total
2026-08-26	22	22	44
2026-08-27	12	18	30
2026-08-28	5	3	8
2026-08-30	0	1	1
2026-08-31	3	2	5
2026-09-01	1	1	2
2026-09-02	0	1	1
2026-09-03	3	0	3
2026-09-04	1	1	2
2026-09-08	21	29	50
2026-09-09	6	1	7
2026-09-10	5	3	8
2026-09-11	4	2	6
2026-09-13	1	2	3
2026-09-14	0	1	1
2026-09-15	1	2	3
2026-09-16	2	0	2
2026-09-17	4	1	5
2026-09-18	3	2	5
2026-09-19	2	1	3
2026-09-20	1	0	1

Total: 97 news + 93 Twitter = 190 records.

The highest daily record volume was observed on September 8, with 50 records, followed by August 26, with 44 records. These two dates together account for 94 of the 190 records in the final dataset. The later sections examine these high-volume periods in greater detail through sentiment, activity and topic analysis.

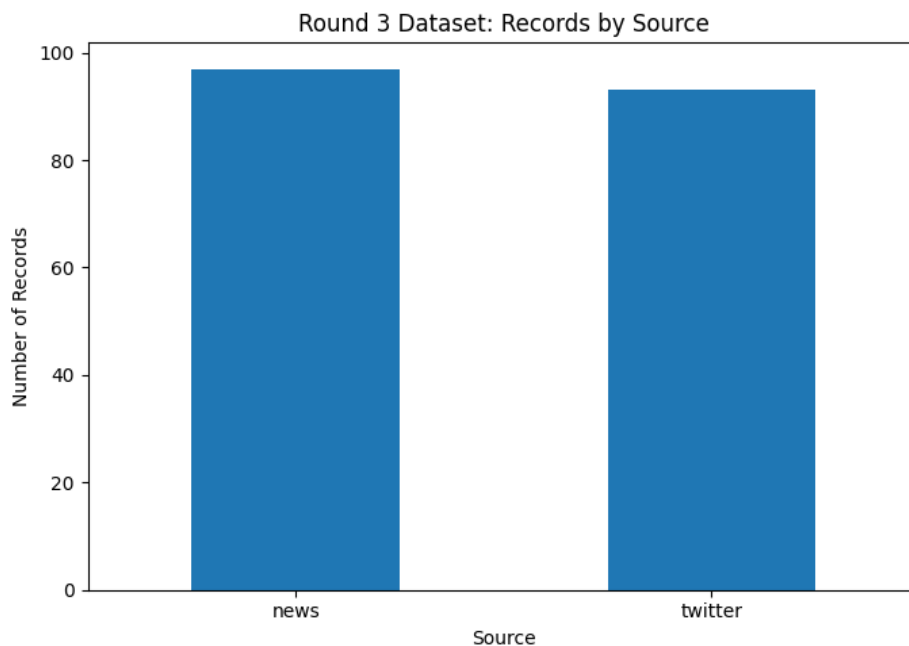
6. Data overview and exploratory analysis

6.1 Data overview

- The final dataset contains 190 records collected from two sources: X (Twitter) and Google News.
- Each record follows the required Round 3 structure containing source, text, timestamp and engagement fields, along with the predicted sentiment and topic labels.
- The dataset contains 93 X (Twitter) records and 97 news records.
- The analysis covers records collected between August 26, 2026 and September 20, 2026.
- The dataset combines social media posts and news records to examine public discussion from both sources.

6.2 Exploratory analysis

Source distribution



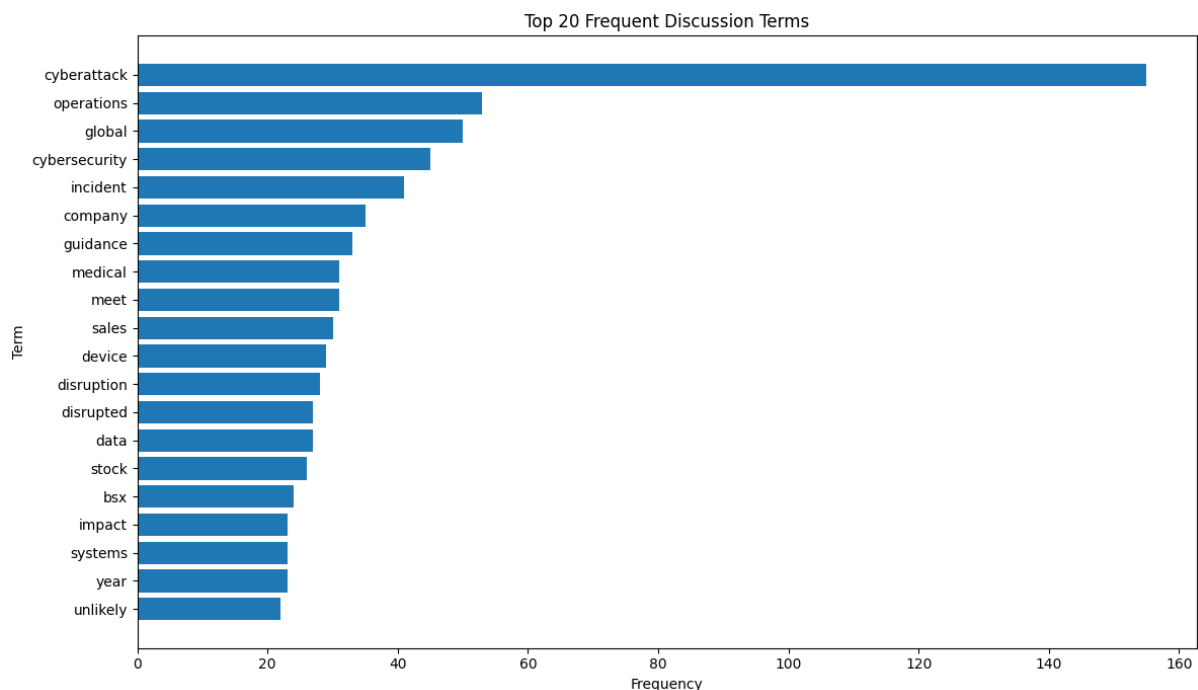
- News contributes 97 records, representing 51.05% of the dataset, while X contributes 93 records, representing 48.95%.
- The two sources are therefore relatively balanced, with only four more news records than X posts.
- This provides representation from both news reporting and social media discussion.
- The source distribution also allows later analysis to compare patterns across the two types of online content.
- Since engagement fields such as likes, comments and shares were available only for X posts, activity analysis is restricted to the X records.

Text-length analysis

Source	Character mean	Character median	Character min	Character max	Word mean	Word median	Word min	Word max
News	91.93	88	59	143	12.70	12	8	23
Twitter	675.08	159	75	12,749	95.37	22	7	1,822

- Across all 190 records, the average text length was 377 characters and 53 words.
- The median was much lower at 100.5 characters and 14 words, showing that most records were relatively short.
- News records were shorter, with an average of about 92 characters and 13 words, consistent with the headline-oriented text collected from the RSS feed.
- X records were longer on average, with about 675 characters and 95 words, although their median was only 159 characters and 22 words.
- The maximum text length was 12,749 characters and 1,822 words, mainly reflecting the presence of a small number of unusually long X records.

Frequent discussion terms



- cyberattack was the most frequent discussion term, appearing 155 times in the dataset.
- Terms such as cybersecurity, incident, operations, disruption and systems show strong focus on the cybersecurity incident and its operational effects.
- Terms including medical, device and sales indicate discussion of both medical-device operations and business impact.
- guidance, stock, impact and year show that financial and company-performance related discussion was also present.
- Overall, the frequent terms show that the collected discussion covered the cybersecurity incident, operational disruption, medical devices and financial impact.

7. NLP methodology — Round 2 model application

7.1 Model introduction

- The NLP models developed and selected during Round 2 were reused for the Round 3 cybersecurity incident dataset without retraining.
- For sentiment classification, the selected model was TF-IDF with Logistic Regression. It classifies text into three categories: Negative, Neutral and Positive.
- The Round 2 sentiment model achieved an accuracy of 0.5734 and a macro F1-score of 0.5730. These results indicate that the model provides a useful classification output but has moderate predictive performance.
- For topic classification, the selected model was TF-IDF with Linear SVM (LinearSVC). It classifies text into four learned categories: Account_Security, Community_Discussion, Feature_Feedback and Technical_Issues.
- The same trained TF-IDF representations and classifiers were applied to the Round 3 text so that the Round 3 analysis remained consistent with the NLP methodology developed in Round 2.

7.2 Sentiment prediction

The trained Round 2 sentiment model was applied to all 190 records in the Round 3 dataset.

Sentiment label	Count	Percentage
Negative	61	32.11%
Neutral	96	50.53%
Positive	33	17.37%
Total	190	100%

- The model classified 96 records as Neutral, making it the largest predicted sentiment category.
- Negative sentiment was predicted for 61 records, representing 32.11% of the dataset.
- Positive sentiment was predicted for 33 records, representing 17.37% of the dataset.
- The distribution shows that the collected discussion was predominantly classified as Neutral, followed by Negative and Positive.
- Since the Round 2 sentiment model had a macro F1-score of 0.5730, these results are treated as model predictions rather than direct measurements of public sentiment.

7.3 Topic prediction

The trained Round 2 topic model was applied to the same 190 Round 3 records without retraining.

Topic category	Count	Percentage
Account_Security	0	0.00%
Community_Discussion	182	95.79%
Feature_Feedback	0	0.00%
Technical_Issues	8	4.21%
Total	190	100%

- The model classified 182 records as Community_Discussion, representing 95.79% of the dataset.
- Eight records were classified as Technical_Issues, representing 4.21% of the dataset.
- No Round 3 records were classified as Account_Security or Feature_Feedback.
- The topic predictions are therefore highly concentrated in the Community_Discussion category, with a small Technical_Issues group.
- The same Round 2 TF-IDF vectorizer was used to transform the Round 3 text, ensuring that the topic model received the same feature representation used during training.

7.4 Interpretation of the topic distribution

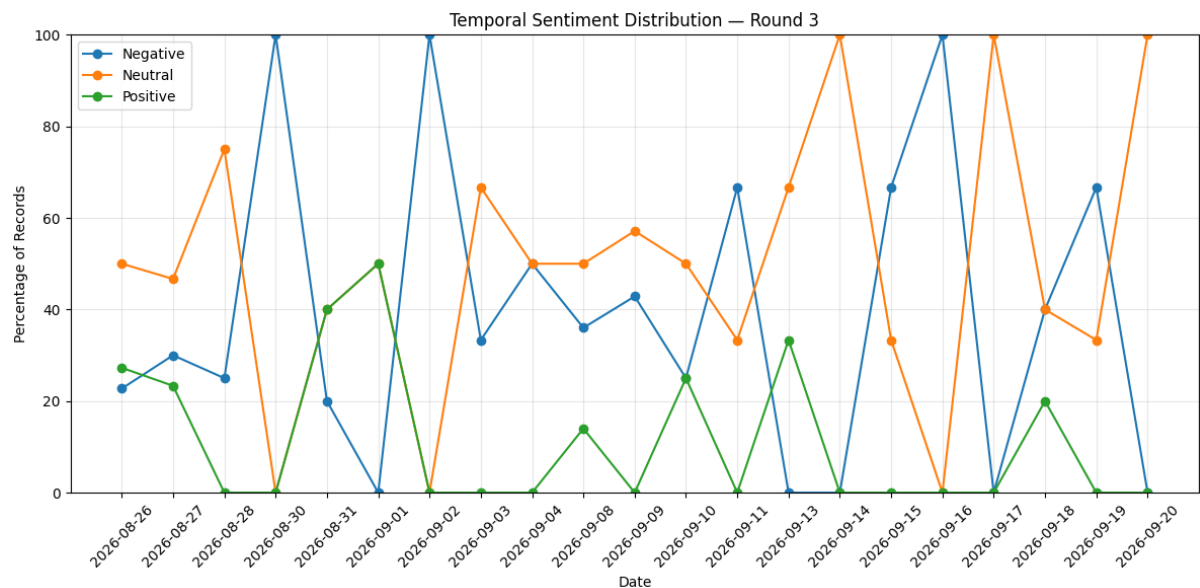
- The strong concentration in Community_Discussion should not be interpreted as proof that 95.79% of the actual public discussion belonged to this topic.
- The Round 2 topic dataset had class imbalance, and the trained model showed a tendency to predict the dominant Community_Discussion class.
- This helps explain why the Round 3 topic predictions are heavily concentrated in one category.
- The eight Technical_Issues predictions are therefore examined separately in the later temporal and event analysis to determine when they occurred and what type of discussion they represented.
- The absence of Account_Security and Feature_Feedback predictions means that the Round 2 classifier did not assign any of the 190 Round 3 records to these learned categories; it does not necessarily mean that these themes were absent from the collected text.

8. Sentiment analysis

8.1 Daily sentiment distribution

The predicted sentiment labels were grouped by date to examine how the sentiment composition changed throughout the analysis period.

Because the labels were generated using the Round 2 TF-IDF + Logistic Regression model, these results represent model-classified sentiment rather than manually verified public sentiment.



The two highest-volume dates were 26 August with 44 records and 8 September with 50 records. These provide the most meaningful basis for comparing sentiment because the remaining dates generally contain substantially fewer observations.

8.2 Key sentiment shifts

Three sentiment shifts were identified from the daily predictions.

1. 26 August to 8 September

This is the most meaningful shift because both dates had high data volume

Sentiment	26 August (%)	8 September (%)	Change (percentage points)
Negative	22.73	36.00	+13.27
Neutral	50.00	50.00	0.00
Positive	27.27	14.00	-13.27

The Negative proportion increased by 13.27 percentage points, while Positive decreased by the same amount. Neutral remained at 50%.

The 26 August period was associated with initial discussion of the cyberattack and operational disruption. By 8 September, the collected discussion included financial impact, company guidance, continued operational effects and recovery-related information. The change in sentiment therefore coincided with a shift in the type of information being discussed.

2. 26 August to 27 August

A smaller shift occurred immediately after the initial high-volume period

- Negative increased from 22.73% to 30.00% (+7.27 percentage points).
- Neutral decreased from 50.00% to 46.67% (-3.33 percentage points).
- Positive decreased from 27.27% to 23.33% (-3.94 percentage points).

The number of records also decreased from 44 to 30. This indicates a somewhat more Negative model-classified composition, but the change is smaller than the 26 August–8 September shift.

3. 8 September to 11 September

A larger percentage change appeared after the 8 September peak.

- Negative increased from 36.00% to 66.67%.
- Neutral decreased from 50.00% to 33.33%.
- Positive decreased from 14.00% to 0.00%.

However, the number of records dropped from 50 on 8 September to only 6 on 11 September. Therefore, this shift should be interpreted more cautiously than the high-volume comparison.

Overall, the strongest observed pattern is the increase in model-classified Negative sentiment and decrease in Positive sentiment between the two major activity periods.

8.3 Possible reasons behind the observed changes

The sentiment changes can be viewed alongside the different stages of the incident represented in the collected data.

1. Initial incident period — 26 August: Discussion focused on the newly reported cyberattack, operational disruption, shipping and order-processing problems, and uncertainty around restoration.
2. Immediate follow-up — 27 August: Discussion continued around the consequences of the incident, including medical devices, services and operational effects. The model showed a small increase in Negative predictions.
3. Financial and recovery period — 8 September: The discussion contained more information about financial impact, company guidance, operational disruption and recovery developments. This period also had the highest total dataset volume of 50 records and the highest X engagement among the major periods.
4. Post-peak period — 9–11 September: The volume decreased considerably, while the proportion of Negative predictions increased. The 11 September value should be treated cautiously because it was based on only six records.

These observations show temporal association between changes in the discussion and changes in model-classified sentiment. They do not establish that a particular incident update directly caused the sentiment change.

8.4 Sentiment model reliability and interpretation

The sentiment model achieved 57.34% accuracy and a macro F1 of 0.5730 in Round 2. The confusion matrix showed notable confusion between Neutral and Negative classes.

Therefore, the Round 3 results should be treated as model-based sentiment signals rather than definitive measurements of public sentiment. Larger-volume dates provide a stronger basis for comparison than low-volume dates.

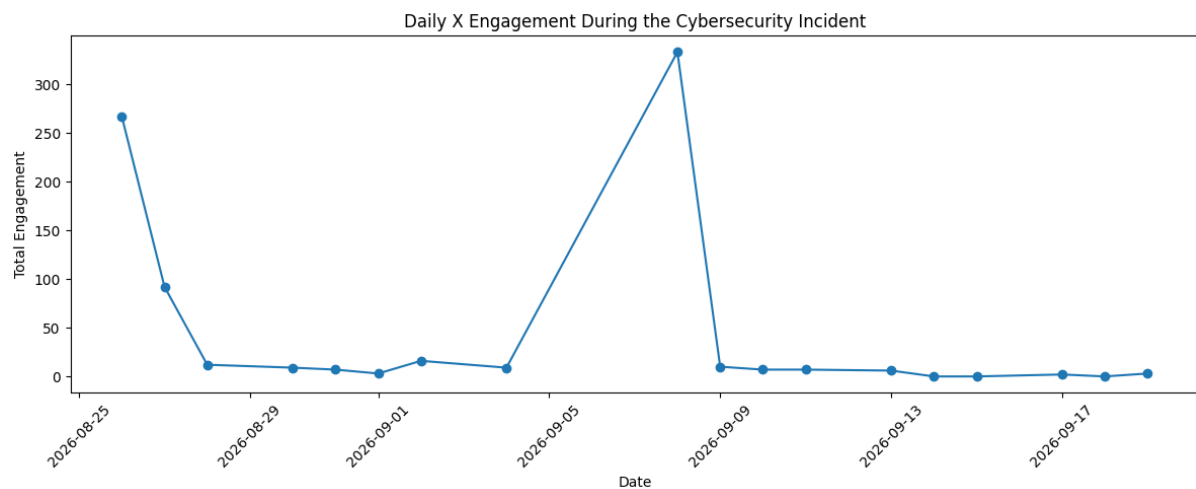
9. Activity analysis

9.1 Daily X engagement

Activity analysis examines how engagement varied across X posts during the cybersecurity incident. Since engagement metrics were available only for X records, this analysis uses likes, comments and shares.

Total engagement was calculated as:

Total engagement = Likes + Comments + Shares



The mean daily X engagement was 43.50, while the highest daily engagement was 333 on 8 September. Thus, the peak was approximately 7.66 times the mean daily engagement. Engagement was concentrated mainly around 26 August and 8 September.

9.2 Highest-engagement posts

The two highest-engagement periods also contained highly engaged individual posts.

Date	Text ID	Total engagement
2026-09-08	2097419837270118678	116
2026-08-26	2092588048303153324	94

The most-engaged post in the dataset was posted on 8 September and received 116 total engagements. Its content discussed the cyberattack stopping Boston Scientific from shipping medical devices to hospitals and clinics and noted that affected systems had not yet been fully restored.

The highest-engagement post from 26 August received 94 total engagements and focused on the disruption caused by the attack, including its effect on Boston Scientific's operations and shipment services.

9.3 Possible explanations for the two engagement spikes

1. 26 August — initial incident and operational disruption

The 26 August period recorded 22 X posts and 267 total engagements, including 84 shares. The discussion focused on the initial cyberattack, disruption to global operations, shipment problems and uncertainty about system restoration. The concentration of shares suggests that information about the newly reported incident was being circulated widely during this period.

2. 8 September — continued disruption and broader impact

The 8 September period recorded 29 X posts and 333 total engagements, the highest in the dataset. Posts discussed continued disruption to medical-device shipments, incomplete system restoration, financial guidance, and possible effects on sales and profit forecasts. The most-engaged post received 116 engagements and focused on the continuing shipping and system-restoration issues.

Overall, the two engagement spikes coincided with periods containing new or consequential information about the incident. These observations show temporal association and do not establish that a particular update directly caused the increase in engagement.

10. Topic and entity analysis

10.1 Key discussion topics and entities

The text-based analysis identified five main discussion themes in the collected records:

Discussion theme	Records	Percentage
Cyberattack & Cybersecurity	189	99.47%
Operational Disruption	90	47.37%
Financial Impact	76	40.00%
Shipping & Medical Devices	57	30.00%
Patient & Service Impact	10	5.26%

The cyberattack and cybersecurity theme was present in almost the entire dataset. Operational disruption, financial impact, and medical-device/shipping issues were also major areas of discussion. The patient and service theme was smaller but included references to pacemakers, LATITUDE and remote monitoring.

The main entities were:

Entity	Records	Percentage
Boston Scientific	187	98.42%
Cybersecurity Incident	171	90.00%
BSX	20	10.53%
Medical Devices	20	10.53%
Pacemaker	2	1.05%
LATITUDE	1	0.53%
Remote Monitoring	1	0.53%

Boston Scientific and the cybersecurity incident were the dominant entities, while BSX and Medical Devices were particularly relevant to the financial and operational discussions.

10.2 Topic evolution across major activity periods

The composition of the discussion changed substantially between the two highest-activity dates.

Theme	26 August	8 September	Change
Cyberattack & Cybersecurity	100.00%	100.00%	0.00 pp
Operational Disruption	79.55%	20.00%	-59.55 pp
Shipping & Medical Devices	40.91%	18.00%	-22.91 pp
Financial Impact	20.45%	82.00%	+61.55 pp
Patient & Service Impact	2.27%	6.00%	+3.73 pp

On 26 August, discussion was mainly focused on the initial cyberattack, operational disruption, order processing, manufacturing and shipping. On 8 September, Financial Impact became the dominant secondary theme, while operational and shipping topics formed a smaller proportion of the discussion.

This represents a change in the composition of the discussion rather than the disappearance of operational or shipping issues.

10.3 Reasons behind the topic changes

The topic changes coincided with the progression of the incident and the availability of new information.

- On 26 August, discussion focused on what had happened and which operations were affected. The main subjects were the cyberattack, global operations, manufacturing, order processing, shipping and medical devices.
- By 8 September, additional information about financial guidance, sales, profit, forecasts, earnings and stock performance had entered the discussion.
- The 8 September data also contained six model-classified Technical_Issues records, compared with none on 26 August.
- These records included system recovery, LATITUDE service restoration, manufacturing, order fulfilment and continued shipping disruption.

Therefore, the discussion moved from the immediate operational effects of the incident toward its broader business, financial and technical consequences

10.4 Round 2 topic model and overall finding

The Round 2 topic model classified 182 of 190 records (95.79%) as `Community_Discussion` and 8 (4.21%) as `Technical_Issues`. No records were classified as `Account_Security` or `Feature_Feedback`. On 26 August, all 44 records were classified as `Community_Discussion`, while on 8 September, 44 of 50 were `Community_Discussion` and 6 were `Technical_Issues`.

The model results therefore support the text-based analysis only as an additional signal, since the Round 2 model strongly favoured `Community_Discussion`. The text-level analysis provides more specific information about the actual subjects being discussed.

Overall, the discussion evolved from:

Cybersecurity incident → operational disruption → shipping and medical-device consequences
→ financial impact → technical and service recovery

The most significant change was the shift from operational-disruption discussion on 26 August toward financial-impact discussion on 8 September.

11. Trigger explanations

This section connects the major changes observed in sentiment, engagement and discussion topics with the incident updates represented in the collected data. The explanations describe temporal associations and do not claim that a specific event directly caused a change.

11.1 Triggers behind sentiment shifts

- 26 August to 8 September — Negative sentiment increased from 22.73% to 36.00%, while Positive decreased from 27.27% to 14.00%. This coincided with the discussion moving from the initial incident toward its continuing operational and financial consequences.
- 26 August to 27 August — Negative sentiment increased from 22.73% to 30.00%. The discussion continued to focus on the effects of the cyberattack, including operational disruption, medical devices and service-related issues.
- 8 September to 11 September — Negative sentiment increased from 36.00% to 66.67%. This coincided with continued discussion of the incident and its consequences, although the 11 September sample contained only six records, so the change should be interpreted cautiously.
- The 8 September period contained more information about financial guidance, sales, profit forecasts and recovery, providing a different context from the initial incident discussion on 26 August.
- These shifts are model-classified patterns. The Round 2 sentiment model had moderate performance, so the observed changes should be treated as sentiment signals rather than definitive measurements of public sentiment.

11.2 Triggers behind engagement spikes

- 26 August — X engagement reached 267 across 22 posts, coinciding with the initial reporting of the cyberattack and its disruption to global operations and shipping.
- The high share count on 26 August, with 84 shares, coincided with widespread discussion and circulation of information about the newly reported incident and uncertainty around restoration.
- 8 September — Engagement reached the dataset maximum of 333 across 29 posts, coinciding with additional information about the continuing effects of the incident.
- Highly engaged posts on 8 September discussed continued medical-device shipping disruption, incomplete system restoration, financial guidance, and possible effects on sales and profit forecasts.
- The increase from 267 engagements on 26 August to 333 on 8 September therefore coincided with a shift from initial incident reporting toward broader operational and financial consequences.

11.3 Triggers behind topic changes

- 26 August — Operational Disruption was present in 79.55% of records, coinciding with the initial disclosure of the cyberattack and reports about operations, manufacturing, order processing and shipping.
- Financial Impact increased from 20.45% on 26 August to 82.00% on 8 September. This coincided with new information about company guidance, sales, profit, forecasts and stock performance.
- Shipping & Medical Devices decreased from 40.91% to 18.00% between the two dates, while shipping disruption continued to appear in the collected records. The change reflects the increasing share of financial discussion rather than the disappearance of the shipping issue.
- Technical_Issues predictions increased from 0% on 26 August to 12% on 8 September. These records included system recovery, LATITUDE service restoration, manufacturing, order fulfilment and shipping-related issues.
- Overall, the topic progression coincided with the incident moving from initial operational disruption toward broader financial, technical and service-related consequences

12. Limitations

- Dataset size — The final Round 3 dataset contains 190 records, with 93 X posts and 97 news records. This limits the amount of data available for detailed temporal and topic analysis.
- Source limitations — Engagement metrics were available only for X posts. The Google News records did not provide likes, comments or shares, so activity analysis was restricted to X.
- Sentiment model limitations — The Round 2 TF-IDF + Logistic Regression model achieved 57.34% accuracy and 0.5730 macro F1. The confusion between Neutral, Negative and Positive classes means that Round 3 sentiment results should be treated as model-based signals.
- Topic model limitations — The Round 2 TF-IDF + Linear SVM model strongly favoured Community_Discussion. In Round 3, 182 of 190 records were classified as Community_Discussion, so text-based keyword and theme analysis was also used to understand the actual discussion.
- Temporal and causal limitations — The analysis identifies patterns and events that coincided with changes in sentiment, engagement and topics. The collected observational data does not establish that a particular incident update directly caused these changes.

13. Conclusion

- The Round 3 analysis successfully combined 93 X posts and 97 Google News records into a validated 190-record dataset using the required nine-field structure.
- The analysis showed that discussion remained strongly centred on the Boston Scientific cybersecurity incident, while the focus changed from operational disruption and shipping issues toward financial and business consequences.
- Sentiment analysis identified changes across the incident timeline, including a higher proportion of model-classified Negative sentiment during the later major activity period on 8 September. These results were interpreted with the limitations of the Round 2 sentiment model in mind.
- Activity analysis identified two major X engagement periods: 26 August with 267 total engagements and 8 September with 333. These spikes coincided with important updates concerning operational disruption, shipping, system recovery and financial impact.
- Overall, the analysis demonstrates how the Round 2 NLP models, combined with text-based topic and entity analysis and temporal engagement analysis, can be used to track how an online cybersecurity discussion changes across different stages of an incident.